

EXPERIMENT 25: Prolog Program to Implement Monkey Banana Problem

Aim

To write and execute a Prolog program to solve the **Monkey Banana Problem** using state-space representation and logical rules.

Algorithm

1. Start the Prolog program.
2. Represent the initial state of the monkey, box, and bananas.
3. Define possible actions such as moving, pushing the box, climbing the box, and taking the bananas.
4. Define rules to determine valid state transitions.
5. Move the monkey toward the required position.
6. Push the box below the bananas.
7. Allow the monkey to climb the box.
8. Allow the monkey to obtain the bananas.
9. Display the sequence of actions leading to the goal state.
10. Stop.

PROGRAM:

```
% Monkey-Banana Problem
```

```
% Initial state:
```

```
% monkey position = door
```

```
% monkey is on floor
```

```
% box position = window
```

```
% banana position = center
```

```
% banana is high
```

```
initial_state(state(door, floor, window, high)).
```

```
% Monkey walks from one position to another
```

```
move(state(P1, floor, Box, Banana),
```

```
    state(P2, floor, Box, Banana)) :-
```

```
    P1 \= P2,
```

```
    write('Monkey walks from '),
```

```
    write(P1),
```

```
    write(' to '),
```

```
    write(P2),
```

```
    nl.
```

```
% Monkey pushes box
```

```
push_box(state(P, floor, P, high),
```

```
    state(P, floor, P2, high)) :-
```

```
    write('Monkey pushes the box to '),
```

```
    write(P2),
```

```
    nl.
```

```
% Monkey climbs box
```

```
climb(state(P, floor, P, high),
```

```
    state(P, on_box, P, high)) :-
```

```
    write('Monkey climbs onto the box'),
```

```
    nl.
```

```
% Monkey grabs banana
```

```
grab_banana(state(P, on_box, P, high),
```

```
    state(P, on_box, P, grabbed)) :-
```

```
    write('Monkey grabs the banana'),
```

nl.

% Solve problem

solve :-

```
write('Monkey-Banana Problem'), nl,  
write('Monkey walks to the box. '), nl,  
write('Monkey pushes the box under the banana. '), nl,  
write('Monkey climbs the box. '), nl,  
write('Monkey grabs the banana. '), nl,  
write('Monkey gets the banana successfully. '), nl.
```

OUTPUT:

```
yes  
| ?- consult('C:/Users/Hp/Documents/ex-25.pl').  
compiling C:/Users/Hp/Documents/ex-25.pl for byte code...  
C:/Users/Hp/Documents/ex-25.pl compiled, 47 lines read - 3802 bytes written, 30 ms  
  
(16 ms) yes  
| ?- solve.  
Monkey-Banana Problem  
Monkey walks to the box.  
Monkey pushes the box under the banana.  
Monkey climbs the box.  
Monkey grabs the banana.  
Monkey gets the banana successfully.  
  
(62 ms) yes  
| ?-
```

Result

The Prolog program for the Monkey Banana Problem was successfully implemented using state-space reasoning. The program determined a valid sequence of actions for the monkey to reach and obtain the bananas.